جامعة خليفة
Khalifa University

Location Name: Lab D100 (Gasifier lab), SAN campus	Description of Task/Activity/Area/Premises etc.: Service visit, servicing, and commissioning of gasifier pilot plant

SN	Step 1	Step 2	Step 3					Step 4					
			Risk Rating					Actions and Implementation			Residual Risk		
	Hazard	Who might be at risk	Probability (P)	Severity (S)	Rating (P X S)	What have you already been doing (List all the Control measure already in place)	Control Measure (Refer to Hierarchy of Control) What further action, if any, is necessary, if so what Action is to be taken by whom and by when?	Action Whom	Action By When	Done %	Probability (P)	Severity (S)	Rating (P X S)
1.	Service visit by experts	Non hazardous	1	1	1	-	No further action required						
2.	Installation of gasifier pilot plant	Non hazardous	1	1	1	-	No further action required						
3.	Commissioning of the installed pilot plant	Personal Injuries & property damage	1	3	3	• Commissioning will be performed under the supervision of experts of knowledge. • All safety precautions, including wearing safety goggles, boots, and gloves will be taken care of. • Each outlet has a proper vent system and no accumulation of gases will take place inside the lab. • Gas detectors have already been installed inside the lab.	No further action required						
4.	Electrical Hazard	Equipment users	2	4	4	• Electrical socket with appropriate circuit breaker will used. • Emergency power-off push buttons are available in the labs. • Electrical system shall be operated by competent	• Manufacturer instructions will be considered and followed. • Appropriate PPE will be used.	Lab users and custodians			1	3	3

						people only. • The electrical units are grounded, plugs fitted with the correctly rated fuse. • Ensure the cables are not in the way of any mobile systems. • After every operation it is the electrical power supply shall be shut off. • Use of only reliable and approved electrical connections is ensured.							
5.	Power Failure	Property Damage	1	2	2	Instrument will shut down upon power failure and there will be no active heating process					1	1	1
	Step 5 :1/11/2022 **Review Date: 2/11/2023**					• Review your assessment to make sure you are still improving, or at least not sliding back. • If there is a significant change in your workplace, remember to check your risk assessment and when necessary amend it.							

Conducted by: Dr. Iqra Reyaz Hamdani
(Postdoctoral Fellow)
(Signature over printed Name)

Endorsed by: Ahammed Basheer Padinjare
(EHS)
(Signature over printed Name)

Approved by:Dr. Ludovic Dumee
(Lab Custodian)
(Signature over printed Name)

PROBABILITY	SEVERITY				
	Insignificant (1)	**Minor (2)**	**Moderate (3)**	**Major (4)**	**Catastrophic (5)**
Rare (1)	**1**	**2**	**3**	**4**	**5**

جامعة خليفة
Khalifa University

Possible (2)	**2**	**4**	**6**	**8**	**10**
Likely (3)	**3**	**6**	**9**	**12**	**15**
Often (4)	**4**	**8**	**12**	**16**	**20**
Frequent/Almost Certain (5)	**5**	**10**	**15**	**20**	**25**

15 - 25	**Extreme Risk**	Absolutely Unacceptably High Activity or industry should not proceed in current form
8 - 12	**High Risk**	Unacceptably High Activity or industry should be modified to include remedial planning and action and be subject to detailed HSE assessment
4 – 6	**Moderate Risk**	Acceptable but must be managed at "As Low As Reasonably Practicable" (ALARP) Activity or industry can operate subject to management and / or modification
1 -3	**Low Risk**	Acceptable without required further action No action required unless escalation of risk is possible

Hierarchy of Control (Controls identified may be a mixture of the hierarchy in order to provide minimum operator exposure)

Elimination	Eliminate the hazard.
Substitution	Provide an alternative that is capable of performing the same task and is safer to use.
Engineering Controls	Provide or construct a physical barrier or guard or technical measures
Administrative Controls	Develop policies, procedures practices and guidelines, in consultation with employees, to mitigate the risk. Provide training, instruction and supervision about the hazard.
Personal Protective Equipment	Personal equipment designed to protect the individual from the hazard.

Risk Score Timeframe	High = 24 hours	Low = 28 days
Extreme = Immediately	Moderate = 14 days	Negligible = Not applicable